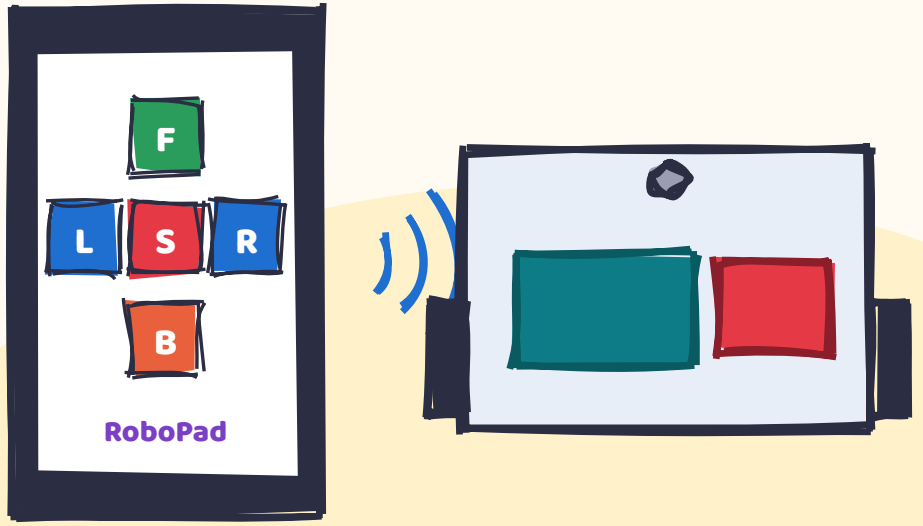
Young Innovators · Class 6

Bluetooth Robot Car with Phone App

Drive a robot from your own phone app

Bluetooth serial | Motor commands | App Inventor | Failsafe



Did you know?
Bluetooth is named after Harald Bluetooth, a Viking king who united Denmark and Norway, just like it unites devices!

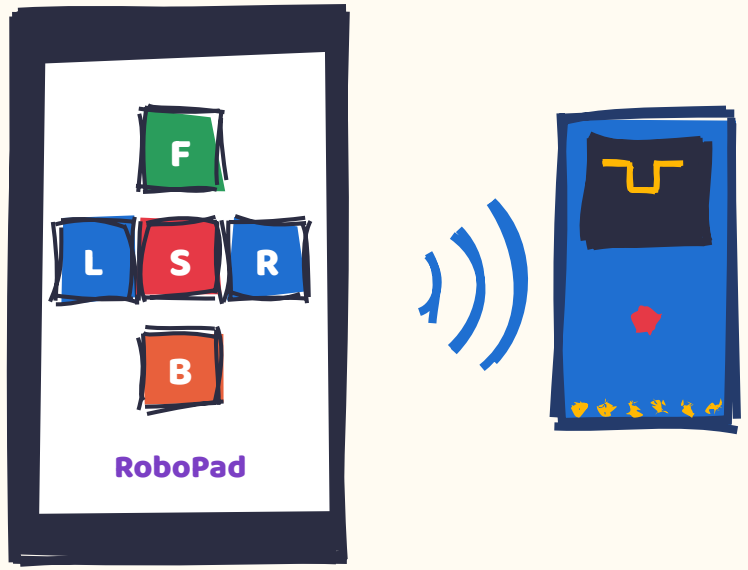
LEARN

Let's think...

How do earbuds talk with no wire?

Invisible wires

Wireless earbuds, a game controller, a smartwatch: they all send data through the air. Your phone can talk to **dozens** of devices. How does it know which one is yours, and how does a button press become a message?



Wi-Fi?

Infrared?

Radio messages!

Today: you build a car that drives with commands sent from **your own phone app** over Bluetooth.

Learning objectives

By the end of this class I can...

1	**Bluetooth:** explain how Bluetooth **pairs** two devices and sends **characters** between them	**Proof:** Serial Monitor shows each letter I send from my phone
2	**Commands:** turn letters like **F, B, L, R, S** into motor movements	**Proof:** my car obeys all 5 commands
3	**App:** build my own **phone controller app** with MIT App Inventor	**Proof:** my app connects and drives the car
4	**Failsafe:** add a **failsafe** that stops the car if the signal is lost	**Proof:** the car stops by itself when I switch Bluetooth off

LEARN

Objective 1 · Bluetooth · video

Video time!



Bluetooth vs WiFi: What's the difference?

PowerCert Animated Videos · about 5 minutes · youtu.be/mPMGRILsOVk

Watch for...

1. What is the **range** of Bluetooth compared with Wi-Fi?
2. What does **pairing** mean?
3. Why does Bluetooth use **less power**?

Scan to watch

After the video: With your partner, list 3 Bluetooth devices at home. Would any of them work better on Wi-Fi?

Objective 1 · Bluetooth

Bluetooth is a wireless serial cable

The idea

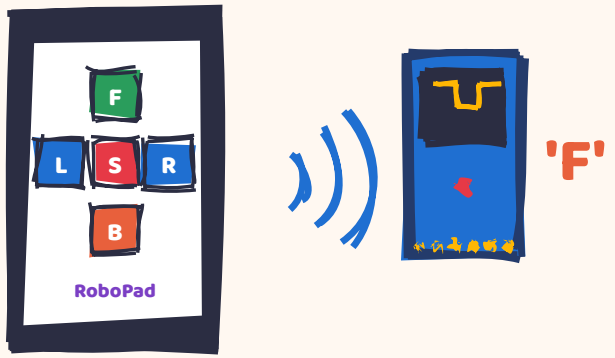
Pairing: the phone and the HC-05 agree to talk, protected by a PIN (usually **1234**).

After that, whatever the phone sends arrives at the Arduino **one character at a time**, exactly like typing in the Serial Monitor. Press **F** on the phone → the Arduino reads **'F'**.

Android only

HC-05 works with Android phones. iPhones use a different kind of Bluetooth (BLE) and need an HM-10 module instead.

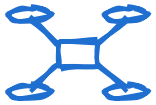
Example: a walkie-talkie



Two walkie-talkies must be on the **same channel** (that is pairing). Then you press a button and speak. The other person hears your words in order. Bluetooth does this with letters instead of voices, about 2.4 billion times faster wobbles!

Real world

Wireless control is everywhere



Drones
Controllers send stick positions many times a second



Game controllers
Buttons travel to the console over Bluetooth



Medical devices
Glucose monitors send readings to a phone



Cars
Phones connect for music, calls and keyless entry

? What should a drone do if it suddenly loses the signal from its controller?

Failsafe
Real drones hover or fly home when the signal is lost. We will add a failsafe too!

Objective 2 · Commands

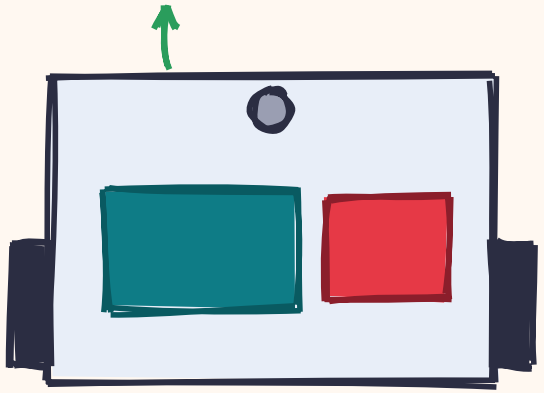
One letter, one movement

The idea

Each wheel has two direction pins. Both wheels forward = **F**. Both back = **B**. Left wheel back + right wheel forward = spin **L**. The opposite = **R**. All off = **S**.

ENA and **ENB** set the speed with PWM (0 to 255).

Example: a rowing boat



L fwd + R fwd = F

L back + R fwd = L

Row with both oars and the boat goes straight. Row with only the right oar, or push it backwards, and the boat spins. A two-wheeled robot steers the same way: it is called **differential drive**.

Objective 4 · Failsafe

Failsafe: no message = stop

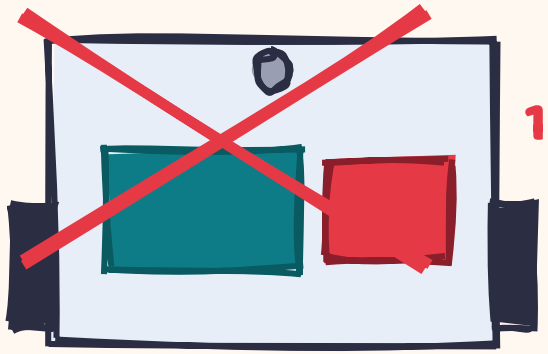
The idea

Make the app send the current letter **every 200 ms** while a button is held (use a **Clock** component).
On the Arduino, add at the end of loop():
if (millis() - lastCmd > 1000) go(0, 0, 0, 0);

If the phone goes out of range or the app crashes, the car stops within 1 second.

Engineering habit
Never trust a wireless link to always work. Safe engineers plan for the moment it fails.

Example: a dead man's switch



Train drivers must keep pressing a pedal or handle. If they faint and let go, the train brakes by itself. Our car does the same when messages stop arriving.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Radio commands
2. Our car kit (per group)
3. Wire the car
4. Pair and say hello
5. Letters become movement
6. Make your own controller app
7. Driving test

MAKE

Class game

Radio commands

How to play

1. One student is the **car**, blindfolded, walking slowly.
2. The **driver** stands across the room and may only shout single letters: **F B L R S**.
3. The class makes 'radio noise' (soft talking): some letters get lost!
4. Guide the car around two chairs and back.
5. Rule: if the car hears **nothing for 3 seconds**, it must stop.

Then discuss

Why did we need a 3-second stop rule? What could happen without it?



Big idea

Short, simple commands are reliable. A failsafe keeps things safe when messages get lost.

MAKE

Build it

Our car kit (per group)



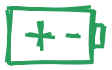
Arduino Uno
Reads commands, drives motors



HC-05 module
Bluetooth receiver



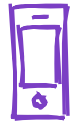
L298N + 2WD chassis
Motor driver, motors, wheels



4×AA pack
Power for the motors



2 LEDs + buzzer
Headlights and horn



Android phone
The controller

Also: 1k and 2k resistors (or two 1k in series) to protect the HC-05's RX pin, plus jumper wires.

Upload tip
Upload the code BEFORE plugging HC-05 into pins 0/1, or use pins 2/3 with SoftwareSerial as we do.

MAKE

Objective 1 · Bluetooth · Build step 1 of 4

Wire the car

Part	Connect to
HC-05 VCC / GND	5V / GND
HC-05 TX	pin 2 (Arduino RX)
HC-05 RX	pin 3 through 1k, with 2k to GND
L298N ENA, IN1, IN2	pins 5, 7, 8
L298N IN3, IN4, ENB	pins 9, 10, 6
LEDs / buzzer	pin 12 / pin 11



Why the resistors?

The HC-05 RX pin wants **3.3 V**. The Arduino sends 5 V. 1k + 2k make a **voltage divider**: 5 × 2 ÷ 3 ≈ 3.3 V.

MAKE

Objective 1 · Bluetooth · Build step 2 of 4

Pair and say hello

```
#include <SoftwareSerial.h>
SoftwareSerial bt(2, 3);      // RX, TX

void setup() {
  Serial.begin(9600);         // to the laptop
  bt.begin(9600);             // to the HC-05
  Serial.println("Waiting for phone...");
}

void loop() {
  if (bt.available()) {       // a letter arrived?
    char c = bt.read();
    Serial.print("Got: ");
    Serial.println(c);
    bt.print("OK ");          // reply to the phone
    bt.println(c);
  }
}
```

Pair

Phone settings → Bluetooth → **HC-05** → PIN **1234**. The red LED blinks slowly once connected in the app.

Test app

Install a free **Serial Bluetooth Terminal** app, connect to HC-05 and type **F**.

Check

Serial Monitor shows **Got: F** and the phone shows **OK F**. Two-way radio!

MAKE

Objective 2 · Commands · Build step 3 of 4

Letters become movement

```ino
#include <SoftwareSerial.h>
SoftwareSerial bt(2, 3);
const int ENA=5, IN1=7, IN2=8, IN3=9, IN4=10, ENB=6;
int speed = 180;                        // 0 to 255
unsigned long lastCmd = 0;

void go(int a, int b, int c, int d) {
  digitalWrite(IN1, a); digitalWrite(IN2, b);
  digitalWrite(IN3, c); digitalWrite(IN4, d);
  analogWrite(ENA, speed); analogWrite(ENB, speed);
}
void setup() {
  for (int p = 5; p <= 12; p++) pinMode(p, OUTPUT);
  bt.begin(9600);
}
void loop() {
  if (bt.available()) {
    char c = bt.read();
    lastCmd = millis();
    if      (c == 'F') go(1, 0, 1, 0);
    else if (c == 'B') go(0, 1, 0, 1);
    else if (c == 'L') go(0, 1, 1, 0);
    else if (c == 'R') go(1, 0, 0, 1);
    else if (c == 'S') go(0, 0, 0, 0);
    else if (c == 'H') tone(11, 800, 200);   // horn
  }
}
```

Wrong way?

If F makes a wheel go backwards, swap that motor's two wires on the L298N.

Speed

Add commands **1** to **9**: speed = digit × 28. Now the phone can change speed!

lastCmd

We record **when** the last letter arrived. We will use it for the failsafe.

Objective 3 · App · Build step 4 of 4

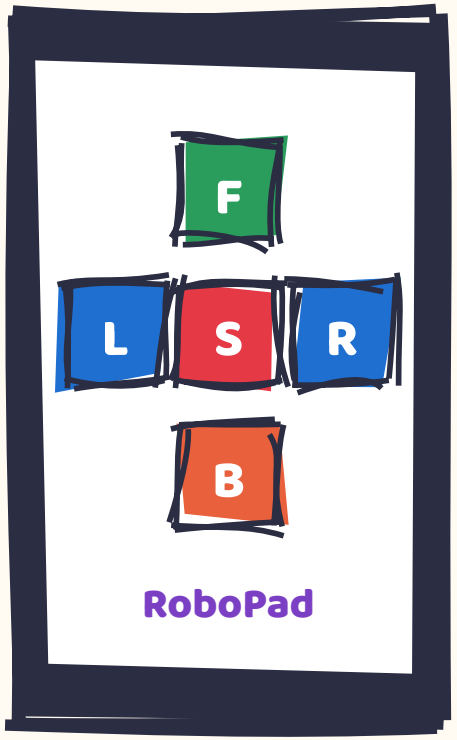
Make your own controller app

MIT App Inventor (ai2.appinventor.mit.edu)

Designer: add a **ListPicker** (Connect), 5 **Buttons** (F, B, L, R, S), and a non-visible **BluetoothClient**.

Blocks:

- ListPicker.BeforePicking → set Elements to BluetoothClient.AddressesAndNames
- ListPicker.AfterPicking → call BluetoothClient.Connect (address = Selection)
- ButtonF.TouchDown → BluetoothClient.SendText **“F”**
- ButtonF.TouchUp → SendText **“S”**



TouchDown / Up
Drive only while your finger is ON the button. Let go = stop. Safer!

MAKE

Test like an engineer

Driving test

Test	Result	Notes
Pair and connect from your app		
F, B, L, R, S all correct		
Drive round 2 chairs and back		
Range: how far until it stops?		____ m
Switch Bluetooth off mid-drive	stops in 1 s?	

Debug checklist

Car jerks or resets? Motors are pulling the battery voltage down. Use fresh batteries and keep the Arduino on its own supply if possible.

Walls and bodies block 2.4 GHz radio. Test range in an open corridor, then through a door.

MAKE

Level up!

Driver challenges

Choose your challenge



Tilt to drive: use App Inventor's AccelerometerSensor to steer by tilting the phone.



Lights: add headlight and horn buttons to your app.



Smart brake: add an ultrasonic sensor that blocks F if something is closer than 15 cm.



Telemetry: send the distance back and show it in the app.

Think like an engineer

Why might sending one letter per command be better than sending whole words like FORWARD?

Boss level

Build an obstacle course race: fastest clean lap wins.

Objective check

Can I really do it? Prove it.

01	What happens during pairing, and why is there a PIN?	○ ○ ○
02	Which pins go HIGH for a left spin? Explain with the rowing-boat idea.	○ ○ ○
03	Show your app connecting and driving the car.	○ ○ ○
04	Switch Bluetooth off while driving. What happens, and which line of code did that?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Driving licence: a partner signs off each objective when they see it working.

Show what you know

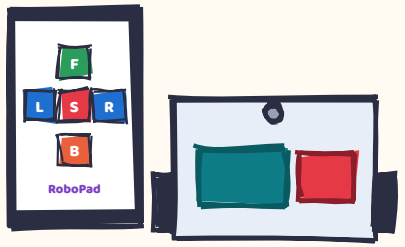
What did we learn?

- [x] Bluetooth pairs devices and sends characters like a wireless serial cable
- [x] One letter per command keeps control simple and fast
- [x] Differential drive steers by running wheels at different directions
- [x] App Inventor buttons send text with BluetoothClient
- [x] A failsafe stops the car when messages stop

Exit ticket: the phone disconnects while the car is driving forward. What should happen, and why?

At home

List the wireless devices in your home. Which ones would need a failsafe?



Next class: Gesture-Controlled Robot Arm!